

PrayogGraph

Hand-Drawn Virtual Physics Labs for Bharat's Classrooms

Zero 3D Engines

Zero Latency

100% Deterministic

The Practical Education Crisis in Bharat



The Infrastructure Gap

- 77% of Indian govt. schools lack functional science labs (ASER 2023).
- Students learn physics via chalkboard diagrams, missing spatial & procedural reasoning.



The "Virtual Lab" Failure

- Existing digital labs rely on heavy 3D/WebGL engines - crashing on Rs.8,000 tablets used in rural schools.
- They are passive click-next videos, not interactive problem-solving environments.



The Teacher Blindspot

- In a class of 50, a teacher cannot monitor individual setups or identify misconceptions in real-time.
- Zero instant corrective feedback mechanism exists.

PrayogGraph: Sketch-to-Simulation Engine

We replaced heavy 3D game engines with a deterministic 2D spatial constraint graph.

01

Hand-Drawn Input

Students sketch circuits & pulleys freehand on a lightweight HTML5 canvas.

02

Spatial Graph Parsing

Engine interprets imprecise strokes, snapping them into a structured logic graph — nodes = components, edges = wires.

03

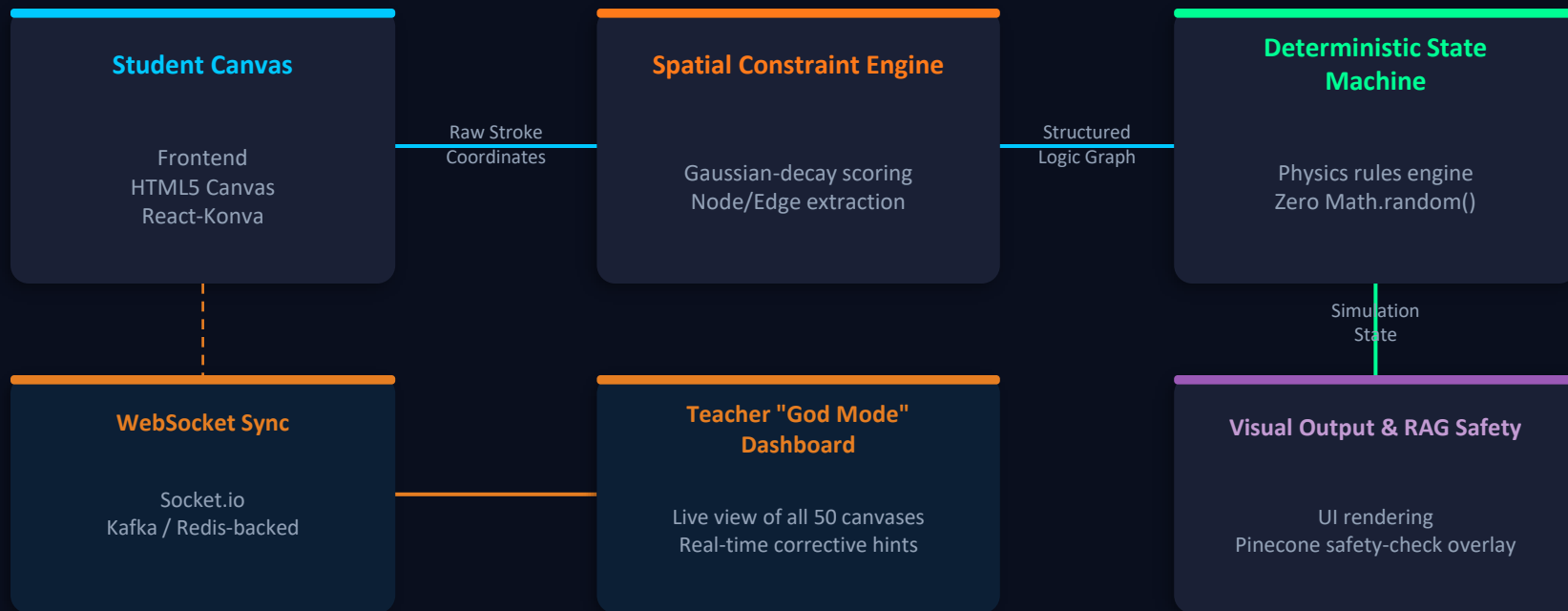
Deterministic Simulation

Zero-randomness state machine computes the physical outcome. Closed circuit = bulb ON. Direct wire = short circuit.

⚡ KEY DIFFERENTIATOR

Runs entirely client-side in <300ms. Works flawlessly on low-end hardware with zero internet dependency after initial load.

System Architecture & Data Flow



Why Our Approach is Technically Superior

✗ The Industry Standard

Fails in Bharat

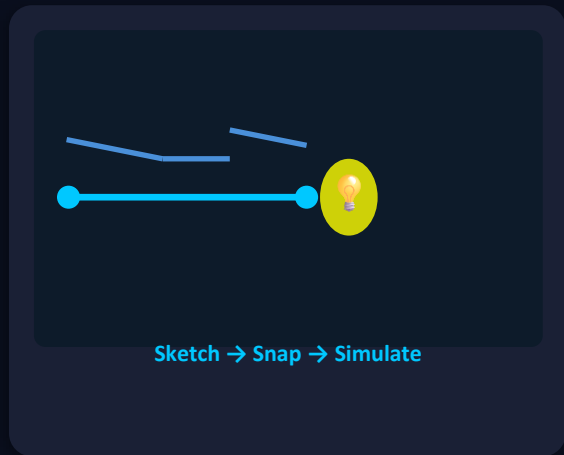
- Relies on Unity/WebGL — High GPU/CPU usage, unusable on budget hardware.
- Uses LLMs for physics simulation — Inherently prone to hallucinations.
- Requires constant 4G connectivity for state syncing.

✓ The PrayogGraph Edge

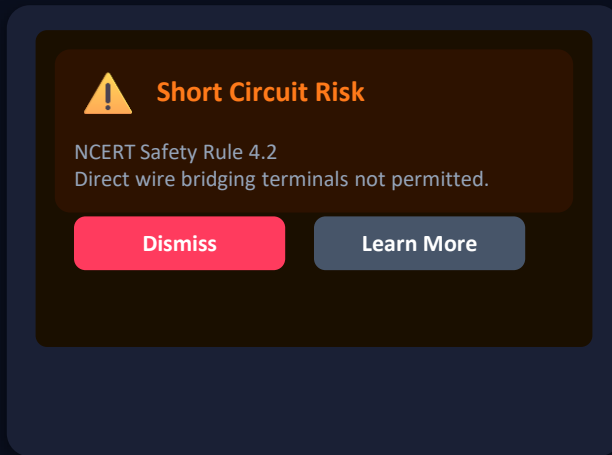
Built for Bharat's Scale

- **Spatial UI Inference:** Repurposes production-grade wireframe layout logic to parse physics diagrams.
- **Mathematically Deterministic:** Same sketch always yields identical output. Verified by automated determinism tests. Zero AI hallucinations.
- **Event-Driven Sync:** Kafka/Redis-backed WebSockets for sub-100ms teacher-student canvas syncing.

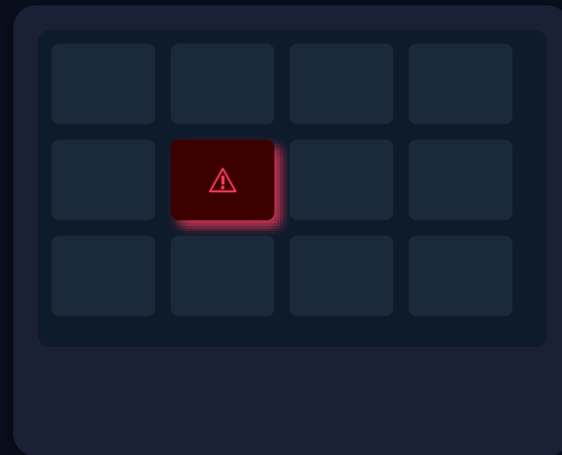
User Experience: Intuitive & Collaborative



Student Canvas



RAG Safety Alert



Teacher Dashboard

Student View: Frictionless sketching. AI hints appear contextually only when stuck for >30 seconds.

Safety First: RAG pipeline (LangChain + Pinecone) cross-references NCERT manuals to block dangerous simulations.

Teacher "God Mode": Real-time grid of all student progress. Teachers can drop in and draw corrective hints live.

Engineered for Performance & Scale



Frontend & Canvas

- Next.js
- TypeScript (Strict)
- React-Konva
- Tailwind CSS



Spatial Engine

- Custom Deterministic Graph Parser
- Gaussian-decay alignment scoring
- Vitest (Determinism Verification)



Backend & Real-Time

- Node.js + Express.js
- WebSockets (Socket.io)
- Inngest (Background Jobs)
- Apache Kafka + Redis



AI & RAG

- LangChain
- Pinecone (Vector DB)
- NCERT Knowledge Base
- Google Gemini (Hint Gen)



Data & Infra

- PostgreSQL
- Redis (Session/Dedup)
- Docker + Vercel
- GitHub Actions (CI/CD)

Impacting the Last Mile of Education

1.4M+

Target Schools

Government schools without functional labs that can instantly deploy PrayogGraph.

<5%

CPU Usage

Our 2D deterministic engine uses a fraction of compute of 3D labs — 50+ concurrent students on one tablet.

₹0

Extra Hardware Cost

Runs on existing government-issued tablets and basic Android phones. No VR or GPUs required.



Pedagogical Shift

Moves assessment from rote memorization to evaluating spatial and procedural problem-solving — the skills students actually need.



SDG Alignment

SDG 4 (Quality Education) — Accessible physics labs for all students, regardless of location.

SDG 10 (Reduced Inequalities) — Zero extra hardware cost bridges the urban-rural divide.

Execution Roadmap

48 Hours

**Hackathon
Prototype**

- HTML5 canvas + spatial constraint graph
- Basic electrical circuits
- Deterministic state machine

Month 1–2

**Alpha
Release**

- RAG safety checks integration
- Teacher WebSocket dashboard
- Pilot: 50 students, 1 school

Month 3–6

**Beta &
Integration**

- Expand to pulleys & optics
- DIKSHA API integration
- 50+ school rollout

Month 6–12

**Bharat
Scale**

- Open-source spatial engine
- 3-district deployment
- Offline-first PWA certification

The Team Behind PrayogGraph



Viraj Sheoran

Lead Backend & Systems Architect

B.Tech CSE, Symbiosis Institute of Technology | CGPA: 9.09 / 10

Why We Can Build This

- Spatial Engine Expertise:** Built D2D — a production-grade deterministic wireframe-to-code compiler using spatial constraint graphs and Gaussian-decay scoring, backed by 55+ automated tests.
- Real-Time Systems:** Architected Meetly (collaborative platform, WebSockets) and built event-driven microservices using Apache Kafka and Redis at InstaSupply.
- AI & RAG Integration:** Developed Learnix — a RAG-based chatbot using LangChain and Pinecone for scoped document retrieval, directly applicable to our NCERT safety pipeline.